

Bank Marketing Campaign Prediction Using Mechanic Learning

Okunbor Promise | Data Analyst Portfolio Project | July 2026

Introduction

Banks offer fixed deposits as an investment product that provides customers with guaranteed returns while giving banks a stable source of funding for lending and other financial activities. The bank uses direct marketing campaigns to promote fixed deposit products. However, contacting every customer is costly, and many customers do not subscribe. As a result, banks need a more effective way to identify customers who are most likely to respond positively before launching future campaigns.

Business Problem

Direct marketing campaigns often require banks to contact a large number of customers, many of whom have little interest in the product. This results in unnecessary marketing expenses and inefficient use of resources.

Project Objective

This project analyzes historical customer and marketing campaign data to identify the factors associated with fixed deposit subscription and develops a predictive model to help the bank target customers more effectively, improve campaign performance, and reduce unnecessary marketing costs. We develop a machine learning classification model that predicts whether a customer will subscribe to a fixed deposit.

Dataset Overview

The project uses the **Bank Marketing Dataset**, which contains historical customer information and marketing campaign records collected during direct marketing campaigns conducted by a Portuguese banking institution.

Item	Value
Number of Records	11,162
Number of Variables	17
Target Variable	Deposit

The dataset includes customer demographic information, financial characteristics, and previous campaign outcomes. Since the target variable (**Deposit**) has two possible outcomes (Yes or No), the problem is treated as a binary classification task.

Feature Categories

Feature Category	Variables
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Customer Information Financial Characteristics	Age, Job, Marital, Education, Balance, Default, Housing, Loan
Current Campaign	Contact, Day, Month, Duration
Previous Campaign	Campaign, Pdays, Previous, Poutcome
Target Variable	Deposit

Methodology

The project was completed in the following stages:

1. Data Cleaning
2. Exploratory Data Analysis (EDA)
3. Feature Engineering
4. Model Development
5. Model Evaluation
6. Business Recommendation

1. Data Cleaning

The dataset was reviewed to ensure it was suitable for exploratory analysis and predictive modeling. No missing values or duplicate records were identified, indicating that the data was complete and reliable.

The data types were verified and found to be appropriate for analysis, with no inconsistencies requiring correction. Some categorical variables contained the value **"unknown,"** representing unavailable customer information rather than missing data. These values were retained as a separate category because they may provide useful insights during analysis and predictive modeling.

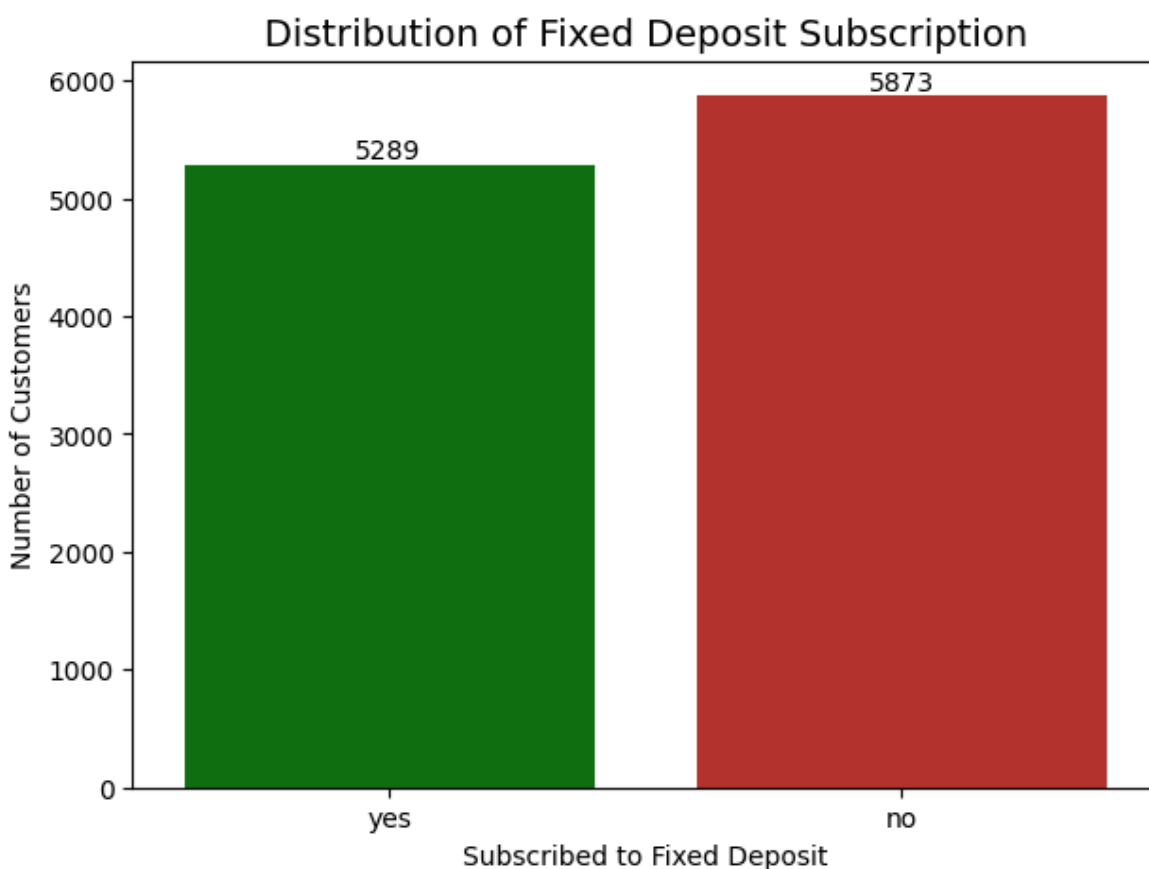
2. Exploratory Data Analysis

Following data cleaning, exploratory data analysis (EDA) was conducted to understand the characteristics of the dataset and identify factors associated with fixed deposit subscription. The analysis examined customer demographics, financial characteristics, marketing campaign variables, and numerical features to uncover patterns that support business decision-making and guide the development of the predictive model.

Understanding the target variable

The target variable was examined to understand the distribution of customers who subscribed to a fixed deposit and those who did not. Understanding the class distribution provides context for subsequent analysis and helps assess whether the dataset is balanced for binary classification.

The results showed that approximately 47% of customers subscribed to a fixed deposit, while 53% did not subscribe. The difference between the two groups is relatively small, which means the historical data contains a good representation of both customers who accepted the offer and those who declined it. This gives us confidence that the analysis can provide reliable insights for future marketing campaigns.



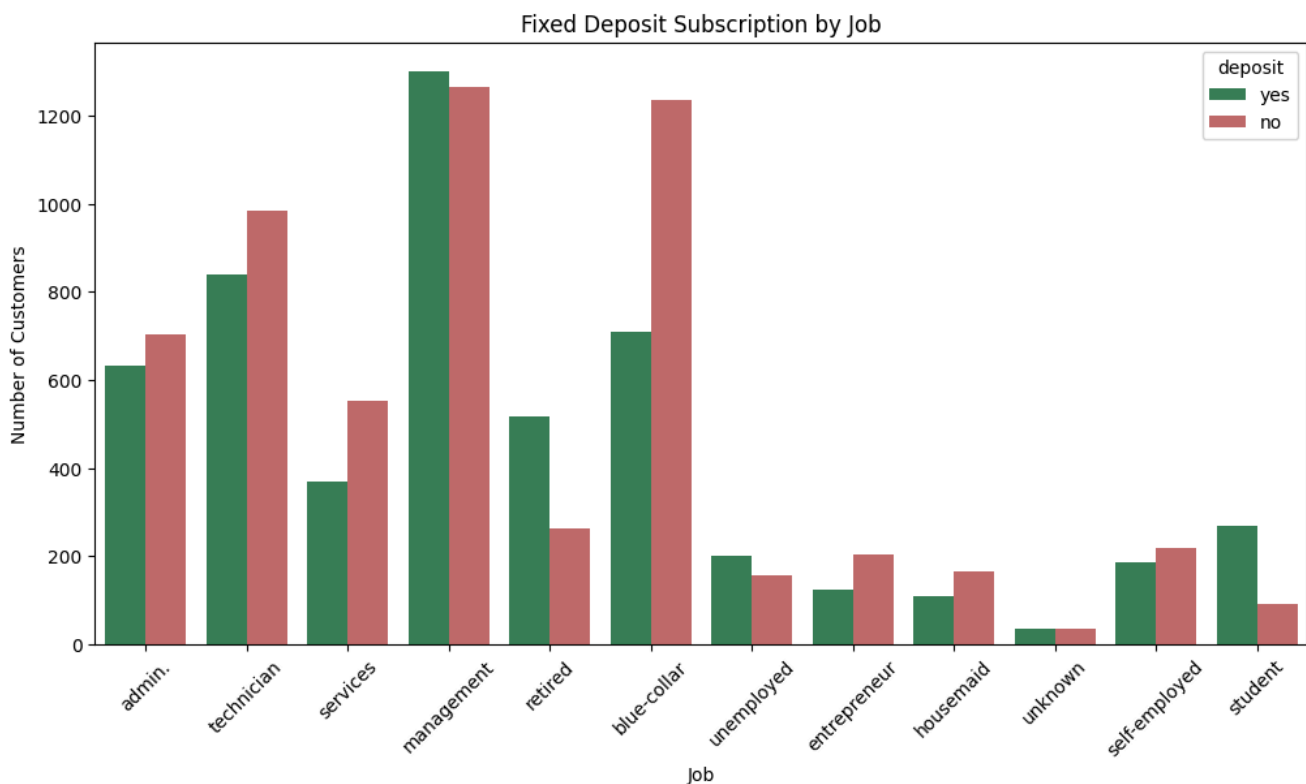
Customer Demographics

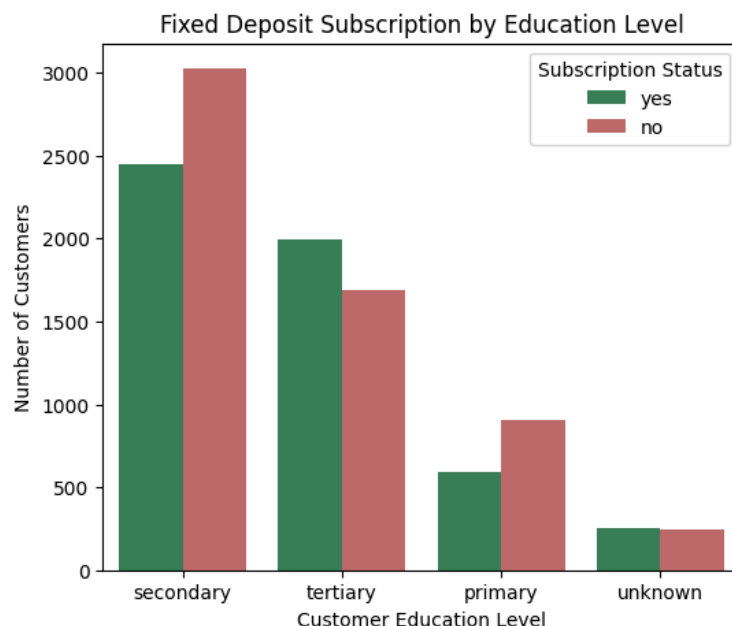
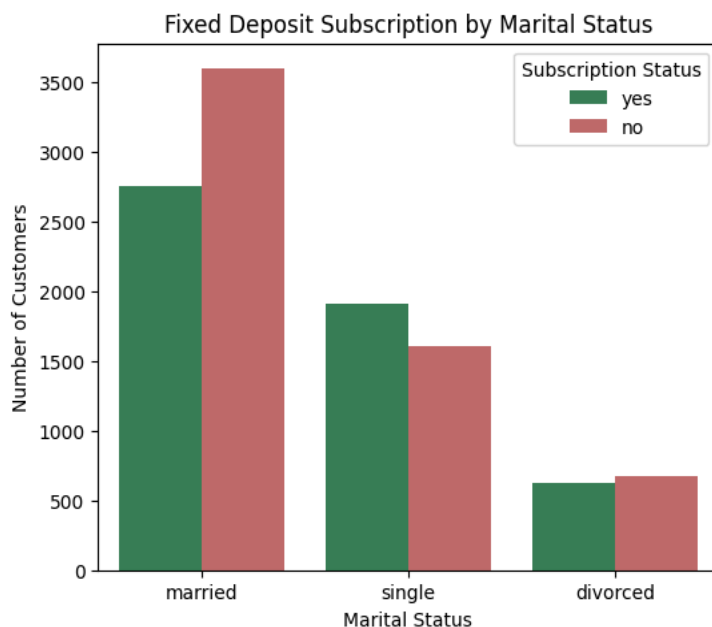
Customer demographic characteristics were analyzed to examine whether occupation, marital status, and education are associated with fixed deposit subscription. Understanding these

characteristics helps identify customer segments that are more likely to respond positively to future marketing campaigns.

The analysis identified clear differences in subscription behaviour across customer demographic groups. Students and retired customers recorded the highest subscription rates, while blue-collar workers recorded the lowest. Single customers were more likely to subscribe than married customers, and customers with tertiary education showed higher subscription rates than those with primary education.

These findings indicate that customer demographic characteristics are associated with fixed deposit subscription. Occupation, marital status, and education level may therefore provide valuable information for customer segmentation and could contribute to improving the predictive performance of the machine learning model.





Financial Characteristics

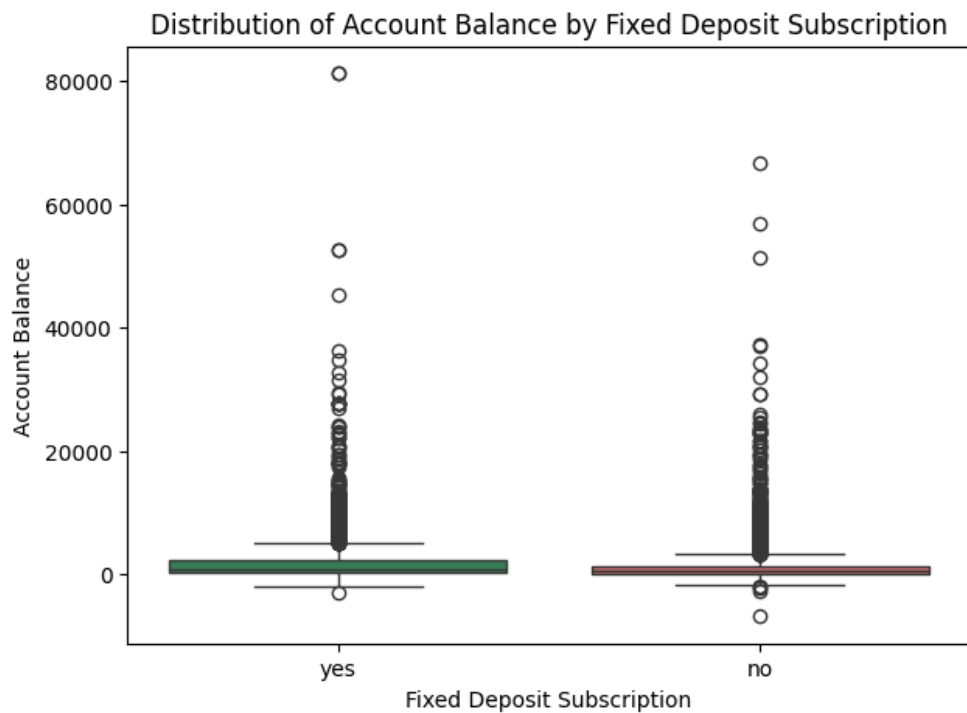
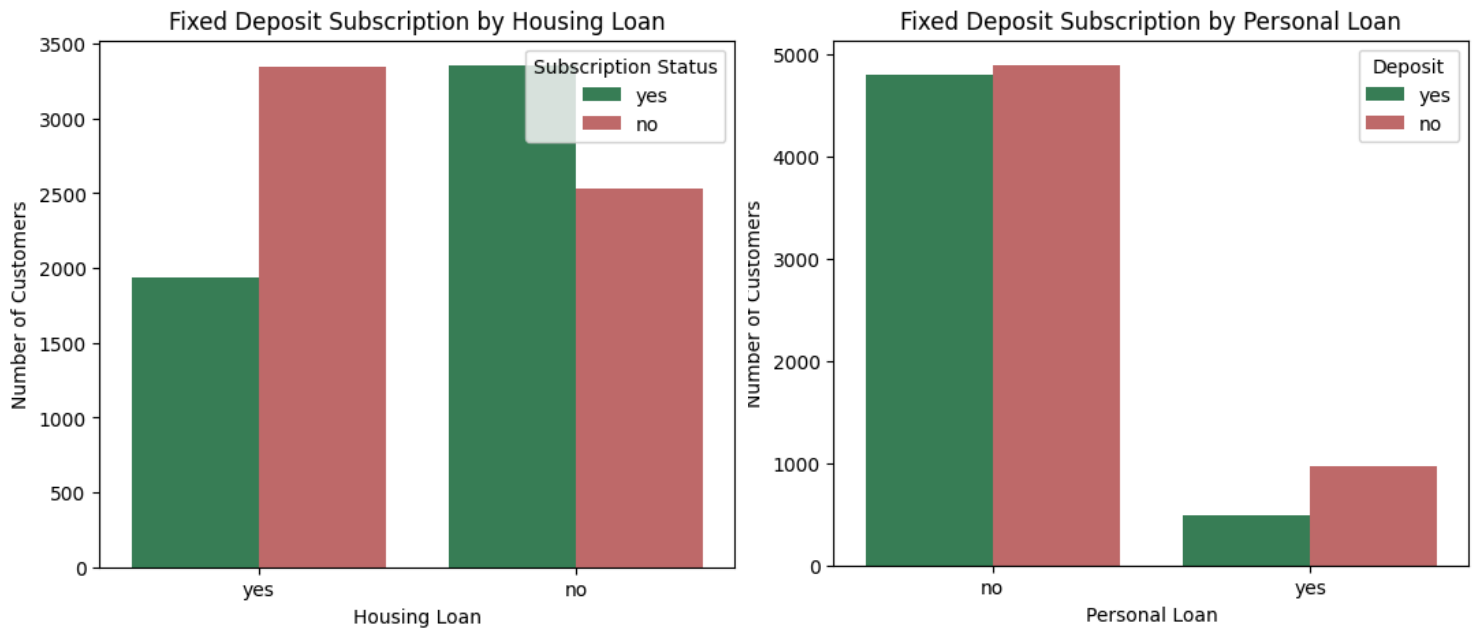
Financial characteristics were analyzed to examine whether customer's financial obligations and account balances are associated with fixed deposit subscription. These factors help identify financial profiles that are more likely to respond positively to future marketing campaigns. However, account balance was not among the top features identified by the final XGBoost model.

The analysis indicates that customers without existing loans were more likely to subscribe to a fixed deposit. Customers without a housing loan recorded a subscription rate of **57.03%**, compared with **36.64%** among customers with a housing loan. A similar pattern was observed for personal loans, where customers without a personal loan recorded a higher subscription rate than those with a personal loan.

Customers who subscribed also tended to have higher account balances, with both the median and average balance exceeding those of non-subscribers, although both groups contained several high-value outliers.

These findings suggest that customers with fewer financial obligations and healthier account balances may have greater financial capacity or willingness to invest in fixed deposit products.

Consequently, loan status and account balance may provide valuable information for customer segmentation and improve the predictive performance of the machine learning model.

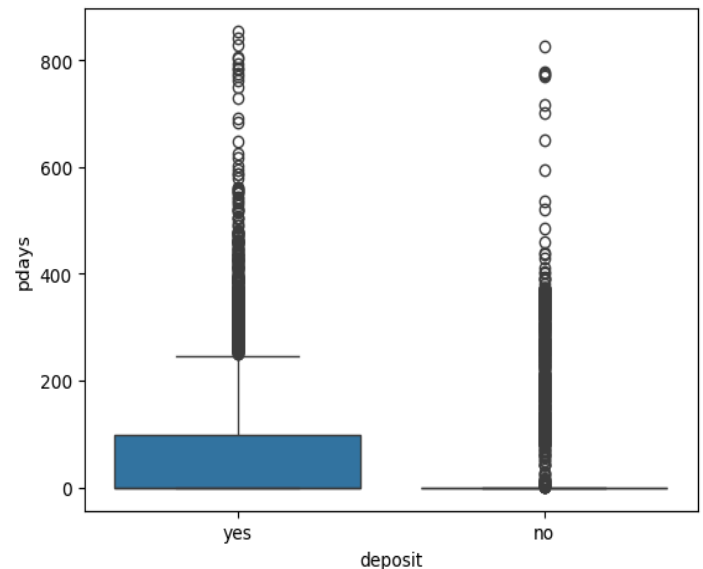
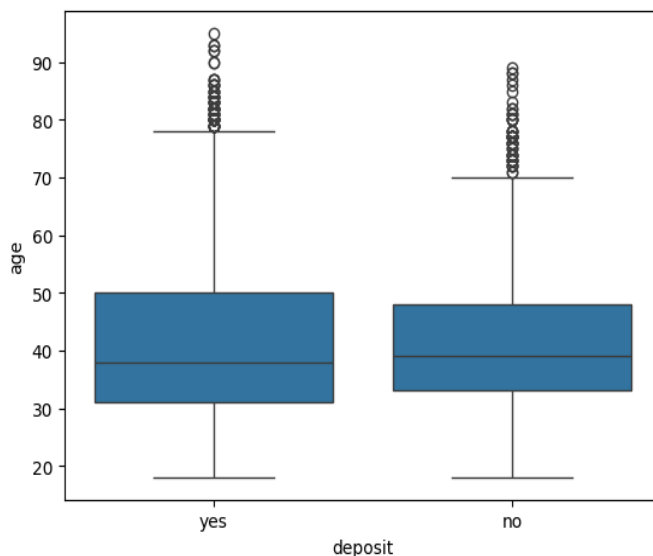
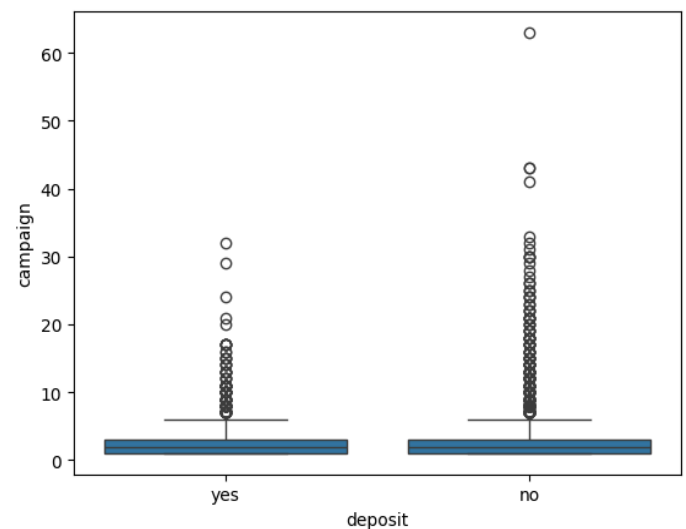
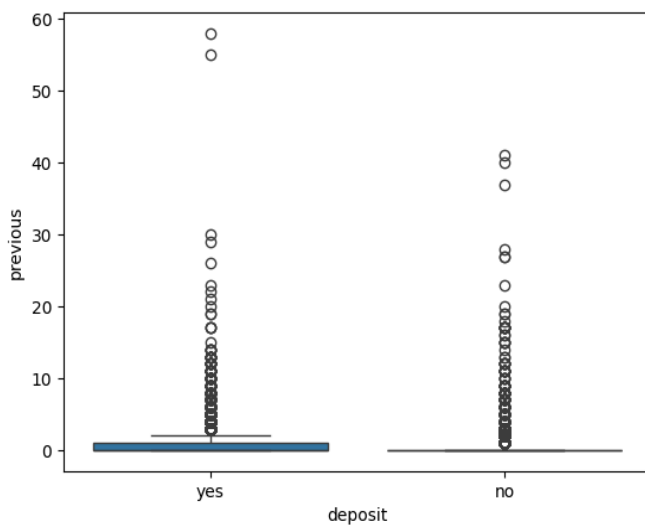


Numerical Variables

The numerical variables were examined to understand their distributions and identify patterns associated with fixed deposit subscription. This analysis provides insight into customer characteristics and previous marketing interactions that may influence the predictive model.

The numerical variables revealed several notable patterns. Customer age showed only small differences between subscribers and non-subscribers, suggesting that age alone is unlikely to strongly influence subscription decisions. Most customers were contacted only a few times during the campaign, while previous campaign variables indicated that many customers had limited or no prior marketing interactions.

The numerical variables helped us better understand customer behaviour and past interactions with the bank. This information improves our ability to identify customers who are more likely to respond positively to future marketing campaigns .



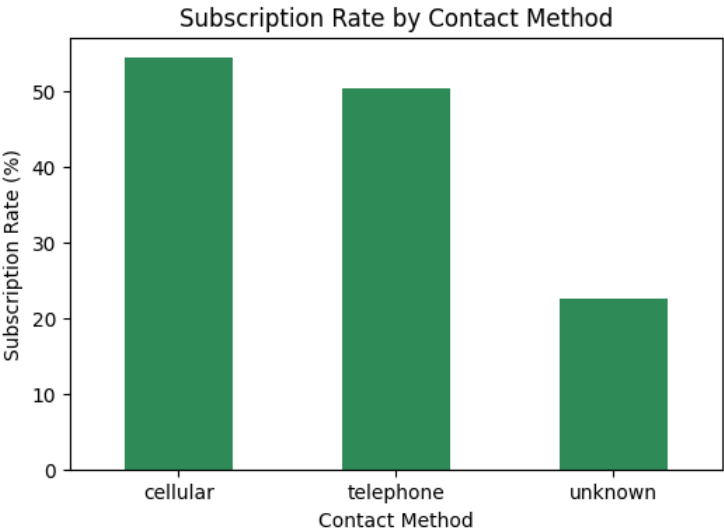
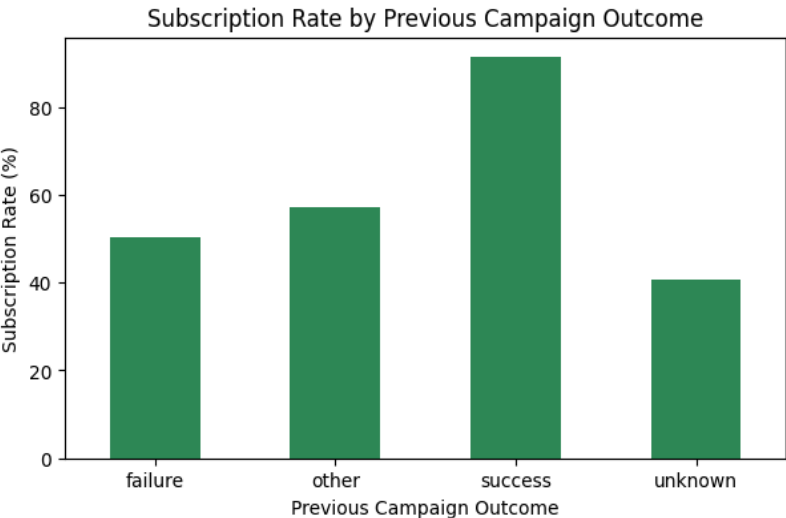
Marketing Campaign Characteristics

Marketing campaign characteristics were also analyzed to evaluate how communication methods and previous campaign outcomes are associated with fixed deposit subscription.

Previous campaign outcomes showed a strong relationship with fixed deposit subscription. Customers who experienced a successful outcome in a previous campaign recorded the highest subscription rate (91.32%), while customers with an unknown previous campaign outcome recorded the lowest subscription rate (40.67%). This may be because customers who responded positively in the past are more familiar with the bank’s products and are therefore more likely to trust and accept future offers.

Contact methods also influenced customer response. Customers contacted via cellular recorded the highest subscription rate, followed by telephone, while those with an unknown contact method had the lowest subscription rate. This suggests that customers respond more easily to mobile communication.

These findings indicate that both communication strategy and previous customer interactions play an important role in fixed deposit subscription. Incorporating these variables into future marketing campaigns and predictive models can help the bank target customers more effectively, improve campaign success rates, and reduce unnecessary marketing costs.



The exploratory data analysis identified several factors associated with fixed deposit subscription. Students, retired customers, and customers with tertiary education recorded higher subscription rates than other demographic groups. Customers without housing or personal loans were also more likely to subscribe, suggesting that financial obligations may influence investment decisions.

Previous marketing interactions emerged as particularly important. Customers who had responded successfully to previous campaigns were substantially more likely to subscribe again, while customers contacted through cellular channels showed higher subscription rates than those reached through other methods.

Feature Engineering

Before model development, the dataset was transformed into a format suitable for machine learning algorithms. Categorical variables such as job, marital status, education, contact type, month, and previous campaign outcome were encoded into numerical representations using one-hot encoding.

The target variable (Deposit) was separated from the input features, and the duration variable was excluded from model training because it becomes available only after a marketing call has ended and could introduce data leakage. These feature engineering steps ensured that the dataset was suitable for training and evaluating the classification models.

Model Development

Four classification models were developed and evaluated to predict whether a customer would subscribe to a fixed deposit. The models included Logistic Regression, Decision Tree, Random Forest, and Extreme Gradient Boosting (XGBoost). Evaluating multiple algorithms made it possible to compare their predictive performance and identify the model that best supports the business objective of improving customer targeting.

Model	Accuracy
Logistic Regression	71.34%
Decision Tree	67.80%

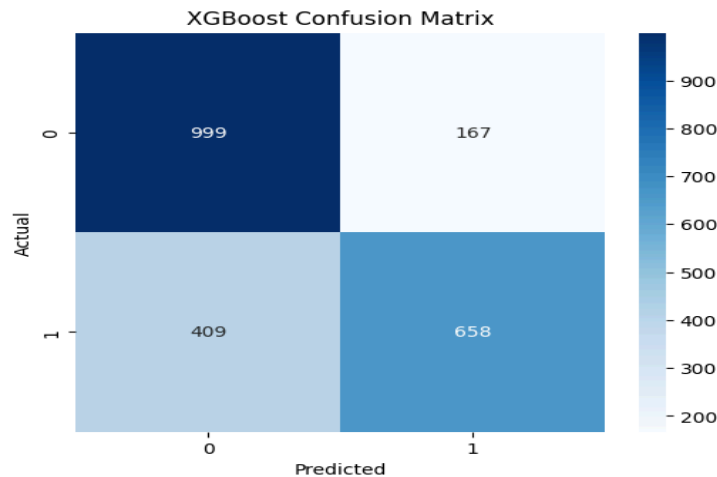
Random Forest	73.44%
XGBoost	74.21%

Model Evaluation

The models showed different levels of performance. XGBoost achieved the highest accuracy (74.21%), followed by Random Forest (73.44%), while Logistic Regression provided a useful baseline model and Decision Tree recorded the lowest performance.

The stronger performance of XGBoost suggests that it was better at identifying the customer characteristics and past campaign patterns associated with fixed deposit subscription. XGBoost was therefore selected as the final model because it provided the most reliable prediction of customers who are likely to subscribe.

By using this model before launching a campaign, the bank can focus on customers with a higher probability of subscribing. This can help improve campaign efficiency, reduce unnecessary contact costs, and make better use of marketing resources.



	precision	recall	f1-score	support
0	0.71	0.86	0.78	1166
1	0.80	0.62	0.70	1067
accuracy			0.74	2233
macro avg	0.75	0.74	0.74	2233
weighted avg	0.75	0.74	0.74	2233

Feature Importance

Feature importance analysis was conducted using the final XGBoost model to identify the variables that contributed most to predicting fixed deposit subscription.

The analysis showed that previous campaign success (`poutcome_success`) was the most influential predictor of subscription. This suggests that customers who responded positively in the past are more likely to trust the bank and subscribe again.

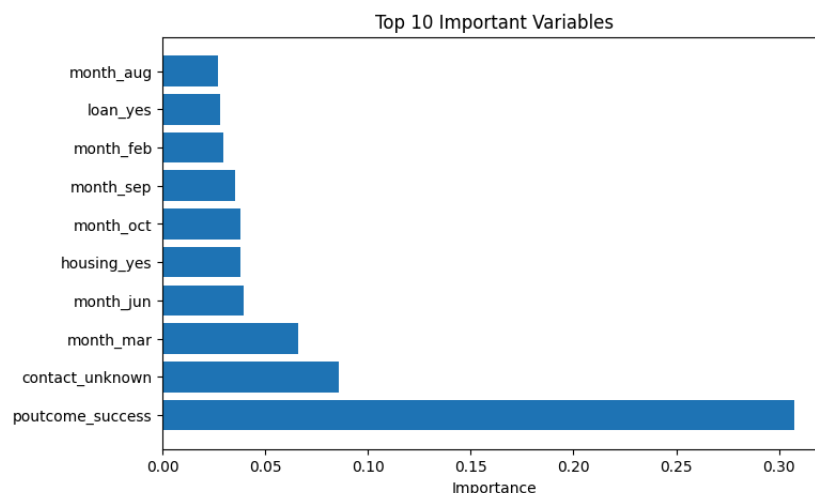
Contact methods also influenced customer response, with cellular contact showing stronger performance. This suggests that customers respond more easily to mobile communication.

Campaign month was another important factor. This may indicate that customer interest varies across different periods of the year, possibly due to seasonal or financial planning patterns.

Housing loan status also contributed to the model. Customers without housing loans may have more available income for investment. Personal loan status was similarly important, as customers with fewer loan obligations may have greater financial flexibility to subscribe to fixed deposits.

These findings show that previous customer engagement, communication strategy, campaign timing, and financial obligations are important factors influencing fixed deposit subscription.

Using these insights can help the bank improve customer targeting, increase campaign effectiveness, and reduce unnecessary marketing costs.



Business Recommendations

Based on the findings of this analysis, the following recommendations are proposed:

1. Prioritize customers with successful previous campaign outcomes

Customers who responded positively in earlier campaigns were significantly more likely to subscribe again. The bank should maintain a list of these customers and contact them first when launching a new fixed deposit campaign or introducing a new investment product. This can improve response rates and reduce the time and cost of reaching less interested customers.

2. Use predictive analytics for customer targeting

When selecting a predictive model for future campaigns, the bank should adopt XGBoost because it delivered the strongest performance among all evaluated models and can help reduce unnecessary marketing efforts by focusing on customers with a higher likelihood of subscription.

3. Focus on effective communication channels

Since customers contacted through cellular channels recorded the highest subscription rates, future campaigns should focus more on mobile communication, as it appears to be the easiest and most effective way to reach customers.

4. Include financial characteristics in customer segmentation

Housing loan status and personal loan status influenced subscription behaviour and should be included in customer segmentation strategies to help the bank identify customers with greater financial flexibility for future fixed deposit campaigns.

5. Plan campaigns during high-response periods

Campaign month was identified as an important factor in the XGBoost model. The bank should further analyze which months are associated with higher subscription activity and consider this information when planning future fixed deposit campaigns.

6. Monitor and update the predictive model regularly

Customer behaviour changes over time, so the model should be retrained periodically using recent campaign data to maintain its effectiveness.

Conclusion

Among the evaluated models, XGBoost achieved the highest predictive performance (74.21%) and was selected as the recommended model for deployment. The model was particularly effective at identifying customers who are likely to subscribe to a fixed deposit and highlighted important business factors such as previous campaign success, communication channel, campaign timing, and customer loan status.

By using XGBoost to support future marketing campaigns, the bank can improve customer targeting, increase campaign efficiency, reduce unnecessary marketing costs, and make more effective data-driven marketing decisions.